

CHEMISTRY BOARD QUESTION PAPER: FEBRUARY 2024**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

- (1) Section A: Q.1 contains ten MCQs of one mark each. Q.2 contains eight very short answer questions of one mark each.
- (2) Section B: Q.3 to Q.14 are twelve short answer questions of two marks each. (Attempt any eight.)
- (3) Section C: Q.15 to Q.26 are twelve short answer questions of three marks each. (Attempt any eight.)
- (4) Section D: Q.27 to Q.31 are five long answer questions of four marks each. (Attempt any three.)
- (5) Use of log table is allowed. Calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) For MCQs, write the correct answer with its alphabet. Only the first attempt will be evaluated.

Given: $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $N_A = 6.022 \times 10^{23}$, $F = 96500 \text{ C}$

SECTION – A**Q.1. Select and write the correct answer for the following MCQs: [10]**

i. Spin-only magnetic moment of Cr^{3+} is —

- (A) 3.742 BM (B) 3.755 BM (C) 3.873 BM (D) 3.633 BM

ii. Linkage present in lactose —

- (A) $\alpha, \beta-1,2$ (B) $\alpha-1,4$ (C) $\beta-1,4$ (D) $\alpha-1,4$

iii. Product of the reaction $(\text{C}_2\text{H}_5\text{COCH}_3 + \text{H}_2/\text{Ni}, \Delta)$ —

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ (B) — (C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

iv. pH of 0.001 M HCl —

- (A) 10 (B) 3 (C) 2 (D) 11

v. Correct structure of sodium hexanitrocobaltate(III) —

- (A) $\text{Na}_3[\text{Co}(\text{NO}_2)_5]$ (B) $\text{Na}_4[\text{Co}(\text{NO}_2)_6]$ (C) $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ (D) $\text{Na}_4[\text{Co}(\text{NO}_2)_5]$

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vi. Number of particles in fcc unit cell —

(A) 1 (B) 2 (C) 3 (D) 4

vii. Monomer used for Teflon —

(A) ϵ -caprolactam (B) vinyl chloride (C) styrene (D) tetrafluoroethene

viii. Vinylic halide —

(A) $R-CHX-R$ (B) $CH_2=CH-X$ (C) — (D) $CH_2=CH-CH_2-X$

ix. Product of hydrolysis of propyne ($HgSO_4/H_2SO_4$) —

(A) methanal (B) ethanal (C) propanal (D) propanone

x. If unit of rate constant is $mol\ dm^{-3}\ s^{-1}$, order of reaction —

(A) 0 (B) 1 (C) 2 (D) 3

Q.2. Answer the following: [8]

i. Name metal nanoparticle used to remove E. coli from water.

ii. Reduction product when ethyl cyanide reacts with sodium and alcohol.

iii. Complete: $CH_3CH_2Cl + AgCN$ (alc, Δ) $\rightarrow$?

iv. Calculate EAN of $[Co(NH_3)_6]^{3+}$.

v. Compounds of Ti^{4+} ions are colourless due to ____.

vi. SI unit of molar conductivity.

vii. Sign convention of work done during gas expansion.

viii. Condition for reverse osmosis.

SECTION – B

(Attempt any EIGHT) [16]

Q.3. Derive expression for maximum work during isothermal reversible expansion from V_1 to V_2 .

Q.4. What are interhalogen compounds? Write reaction of chlorine with dry slaked lime.

Q.5. What is nanomaterial? Write reaction involved in sol–gel hydrolysis.

Q.6. Write classification of proteins with an example.

Q.7. Calculate time to deposit 2.4 g Cu using 2.03 A current. (Atomic mass 63.5 g/mol)

Q.8. Why are amines basic? Among dimethylamine ($pK_b = 3.27$) and diethylamine ($pK_b = 3.0$), which is stronger base?

Q.9. Explain buffer action of sodium acetate–acetic acid buffer.



- Q.10. Preparation of (i) diethyl ether (ii) ethyl cyanide from ethyl bromide.
- Q.11. Henry's constant for $\text{CH}_3\text{Br} = 0.159 \text{ mol dm}^{-3} \text{ bar}^{-1}$ at 25°C . Calculate solubility at 0.164 bar.
- Q.12. Write structure and monomer name of (i) Nylon-6 (ii) Natural rubber.
- Q.13. Define lanthanide contraction. Write equation when acidified $\text{K}_2\text{Cr}_2\text{O}_7$ reacts with H_2S .
- Q.14. Derive relation between molar mass, density and edge length of unit cell.

SECTION – C**(Attempt any EIGHT) [24]**

- Q.15. What is osmotic pressure? How to determine molar mass using osmotic pressure?
- Q.16. Write reactions for: (i) Rosenmund reduction (ii) Gattermann–Koch formylation (iii) Cannizzaro reaction (methanal).
- Q.17. Calculate $\Delta H^\circ_{\text{comb}}$ of methane given $\Delta H_f^\circ(\text{CH}_4) = -74.8$, $\text{CO}_2 = -393.5$, $\text{H}_2\text{O} = -285.8$ kJ/mol.
- Q.18. Action on ethyl bromide: (i) AgNO_2 (ii) Mg /dry ether (iii) alcoholic NaOH .
- Q.19. For $\text{A} + \text{B} \rightarrow \text{P}$, doubling $[\text{B}]$ doubles rate; tripling $[\text{A}]$ and doubling $[\text{B}]$ increases rate 6 times. Find rate law.
- Q.20. Arrange in increasing:
- $\text{HOCl} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$ (acidic strength)
 - MF , MCl , MBr , MI (ionic character)
 - HF , HCl , HBr , HI (thermal stability)
- Q.21. Explain Wolff-Kishner reduction. Prepare propanone using ethanoyl chloride and dimethylcadmium.
- Q.22. Postulates of Werner's theory. Name a hexadentate ligand.
- Q.23. Define electrochemical series; write two applications.
- Q.24. Identify A, B, C (chain reaction) and rewrite reactions.
- Q.25. Define acids & bases (Brønsted–Lowry). Derive relation between pH and pOH.
- Q.26. Preparation of potassium dichromate from chromite ore.

SECTION – D**(Attempt any THREE) [12]**

- Q.27. Convert: (i) acetaldehyde $\rightarrow$ isopropyl alcohol
(ii) cumene $\rightarrow$ phenol
(iii) anisole $\rightarrow$ phenol. Write two uses of neon.



Q.28. Define: (i) extensive vs intensive properties (ii) isobaric vs adiabatic processes.

What are enzymes? Write atomic numbers of transuranium elements.

Q.29. Predict cubic lattice type for edge 400 pm, density 6.25 g/mL (atomic mass 60).

Define nanoscience. Write reaction for preparation of polyacrylonitrile.

Q.30. Derive $t_{1/2}$ – k relation for first-order reaction.

Write net cell reaction for discharging of lead accumulator.

Draw structure of peroxymonosulphuric acid.

Q.31. Number of unpaired electrons & geometry of:

(a) $[\text{NiCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$

Convert: (a) ethanenitrile $\rightarrow$ ethanal (b) cyclohexane $\rightarrow$ adipic acid.

